Swapnil Kumar Goel

Image Processing Engineer | Bengaluru, India
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Summary

Image Processing Engineer, worked with hyperspectral and SAR satellite imagery: georeferencing, band alignment, mosaicking, image registration, SAR autofocus and automated QA/QC. Built Python processing pipelines, Web GIS and QGIS plugins.

Experience

Pixxel Bengaluru, Karnataka
Associate Image Processing Engineer *Nov 2025 – Present*
- Developed image processing workflows for georeferencing, band alignment, mosaic generation and automated AOI clipping.
- Built a Web GIS application and an error-monitoring dashboard for tracking hyperspectral data processing.
- Automated quality assurance and quality control (QA/QC) checks for processed satellite image products.
- Developed two QGIS plugins for hyperspectral data analysis.

GalaxEye Space Bengaluru, Karnataka
Signal Processing Intern – Synthetic Aperture Radar (SAR) *May 2025 – Nov 2025*
- Identified and fixed a bug in the processing pipeline which was resulting in unbalanced SAR image across swath.
- Performed manual georeferencing of SAR Images using GCPs (Ground Control Points), later automated it using LoFTR.
- Automated Image QA that reports along and across-track balance, positional accuracy (CE90), Sharpness (LSF/FWHM).

National Institute of Science Education and Research (NISER) Bhubaneswar, Odisha
Summer Research Intern – School of Earth and Planetary Sciences *May 2023 – Jul 2023*
- Georeferenced Chandrayaan-2 Imaging Infrared Spectrometer (IIRS) hyperspectral images for accurate lunar mapping.
- Aligned spatial coordinates using the SPICE toolkit and Python; computed incidence, emission and phase angles.
- Validated results against orbiter images and known lunar surface features.

Projects

GeoX – Sentinel-2 & COG Viewer | *Python, FastAPI, rio-tiler, MapLibre* – geox-goel.vercel.app *2026*
- STAC search, band composites, NDVI/NDWI, pixel spectra, profiles, region stats, raster calculator; serverless COG tiles.

HORIZON – Sentinel-2 Footprint Explorer | *JavaScript, MapLibre, STAC* – horizon-goel.vercel.app *2026*
- All Sentinel-2 L2A footprints worldwide (12,000/day) by date; search, filters, previews, one-click open in GeoX.

Education

Indian Institute of Technology (Indian School of Mines), Dhanbad Dhanbad, Jharkhand
M.Sc.Tech. in Applied Geophysics – GPA: 8.27/10 *Aug 2022 – May 2025*
- **Thesis:** 3D Seismic Volume Denoising Using Advanced Deep Learning Techniques.

Mahatma Gandhi Kashi Vidyapith (MGKVP) Varanasi, Uttar Pradesh
B.Sc. in Mathematics and Physics – GPA: 8.05/10 *Aug 2019 – May 2022*

Skills

Remote Sensing & GIS: Image Registration, Mosaicking, Geocoding, QA/QC, QGIS, SPICE Toolkit (SpiceyPy), STAC

Image Processing & Data: NumPy, SciPy, OpenCV, scikit-image, GDAL, Rasterio, Pandas, Matplotlib

Machine Learning: PyTorch, TensorFlow, scikit-learn

Programming: Python, MATLAB, Java, SQL (MySQL), JavaScript, HTML, CSS, p5.js

Tools: Git, Docker, Linux, VS Code, Jupyter Notebook, MS Office, Google Workspace

Soft Skills: Problem Solving, Quick Learner, Aptitude and Logical Reasoning

Certifications

MathWorks: Image Processing for Engineering and Science Specialization – Certificate *Jan 2025*

Imperial College London: Mathematics for Machine Learning Specialization – Certificate *Mar 2024*

Achievements

1st Place, Terraprospect Geophysics Hackathon (Team OpendTectives) *Oct 2024*

All India Rank 4, GATE Geophysics 2025 *Feb 2025*

All India Rank 465, IIT-JAM Physics 2022 *Feb 2022*

Qualified, JEE Main 2018 *Apr 2018*

Extracurriculars & Hobbies

Silver Medal (2nd), Carrom – Diamond Day Sports Championship 2025 *Apr 2025*

Gold Medal (1st), Badminton – Intra-hostel Sports Championship 2024 *Mar 2024*

Also casually play chess and table tennis; enjoy brain teasers – Sudoku, Hidato, Rubik's Cube.